OpenSource CDS | Printing Guide

Self-Printing Documentation

Project name: BEP PME-A06

Version: v1.0

Date: May 31, 2026

This document contains the recommended print settings and part overview for printing all required components of the OpenSource CDS.

General Print Requirements/Settings

The parts in this project are designed for standard FDM/FFF 3D printing. All parts can be printed using a single 1 kg roll of filament. A dark, low reflective filament colour is recommended for best results.
Unless stated otherwise for a specific part, use the following settings as a starting point.

Printer type	FDM / FFF 3D printer
Material	PLA recommended, PETG optional for stronger parts
Nozzle diameter	0.4 mm
Layer height	0.20 mm
Wall count	2 walls
Top / bottom layers	4 top layers, 3 bottom layers
Infill	20%
Infill pattern	Gyroid recommended, cubic also sufficient
Supports	Always on build plate only and add supports only where specified per part. Standard threshold angle at 30°
Build plate adhesion	Use a brim only where specified per part
Print orientation	Use the orientation shown per part
Printing speed	Use the standard speed settings from your dedicated printer, unless stated otherwise
Units	Millimetres

Tolerances and Fit

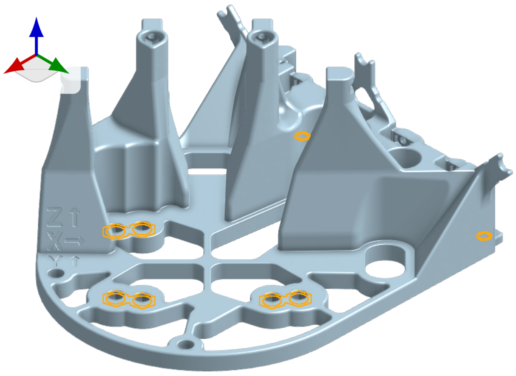
Some parts may need light sanding, cleaning, or adjustment after printing. If parts are too tight, it is recommended to slightly reduce flow rate or increase horizontal expansion compensation in the slicer. Changing dimensions using CAD-software is also a possibility.
If parts are too loose, increase the wall count or check the printer calibration. Again, changing dimensions is a great alternative if only slicer setting changes are not sufficient.

Parts Overview

The following 3D printed parts are required for the full project.

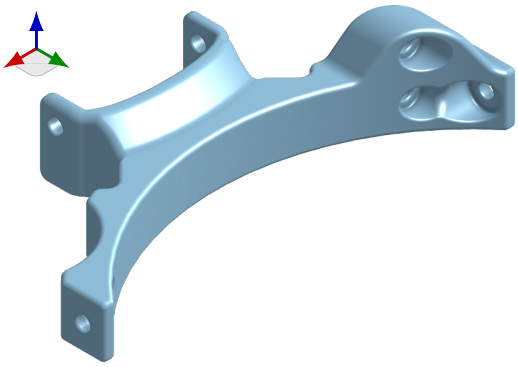
Part	Quantity
BASE_base	1
BASE_bridge	1
BASE_electro	1
BASE_electro_lid	1
BASE_light_mount_base	1
BASE_light_mount_swivel	1
BASE_ruler	1
BASE_top_cap	1
BASE_switch_housing	1
BLOCKS_camera_block	1
BLOCKS_camera_cap	1
BLOCKS_camera_nut_holder	1
BLOCKS_cap	2
BLOCKS_detector_block	1
BLOCKS_detector_clamp	1
BLOCKS_detector_q_slider	1
BLOCKS_detector_side_slider	1
BLOCKS_f1_block	1
BLOCKS_f2_block	1
BLOCKS_guide_spacer	1
BLOCKS_laser_block	1
BLOCKS_laser_cap	1
BLOCKS_laser_spacer	1
BLOCKS_lens_clamp	2
BLOCKS_mirror_block	2
BLOCKS_mirror_clamp_bottom	2
BLOCKS_mirror_clamp_top	1
BLOCKS_redfilter_clamp_top	1
BLOCKS_spacer_0.4	1
BLOCKS_spacer_0.8	1
BLOCKS_spacer_1.2	1
BLOCKS_spacer_1.6	1
BLOCKS_spacer_2.0	1
BLOCKS_spacer_2.4	1
BLOCKS_spacer_2.8	1
BLOCKS_spacer_3.2	1
BLOCKS_spacer_box	1
CALIBRATION_table_calibrator	1
STAGE_actuator_nut_tool	1
STAGE_actuator_ring_tool	1
STAGE_gear	3
STAGE_main_base	1
STAGE_main_body	1
STAGE_moving_platform	1
STAGE_small_gear	3

BASE_base



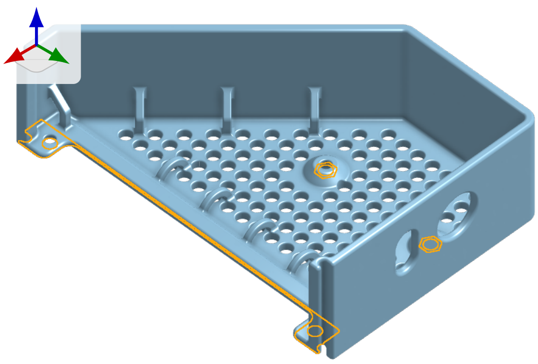
Quantity	1
Wall count	Preferably 3
Supports	On highlighted faces only - Style: normal
Speed	Standard
Notes	-

BASE_bridge



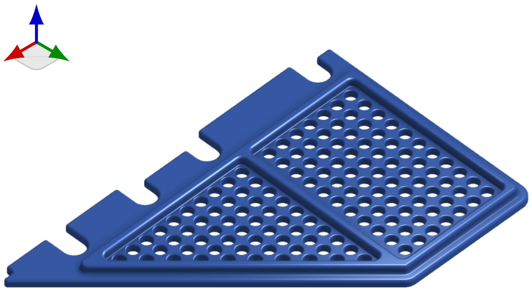
Quantity	1
Wall count	2
Supports	-
Speed	Standard
Notes	-

BASE_electro



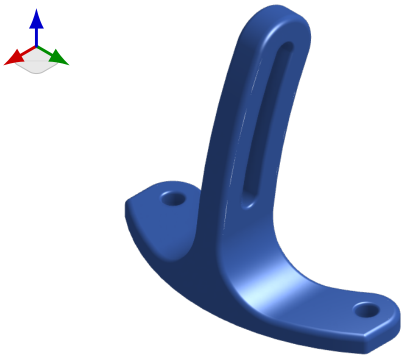
Quantity	1
Wall count	2
Supports	On highlighted faces only - Style: normal
Speed	Standard
Notes	-

BASE_electro_lid



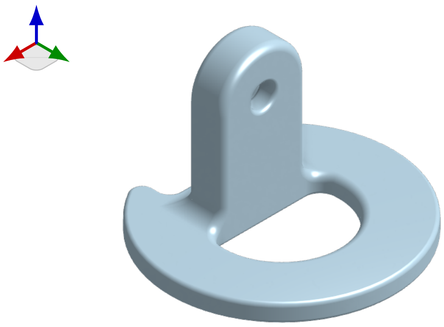
Quantity	1
Wall count	2
Supports	-
Speed	Standard
Notes	-

BASE_light_mount_base



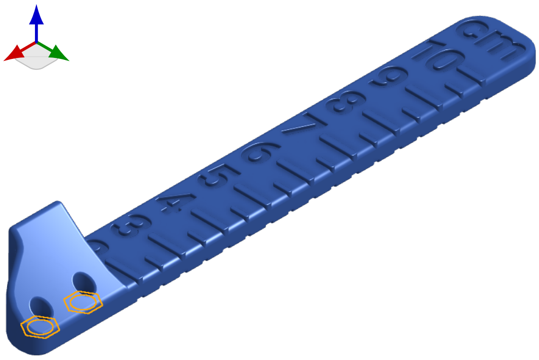
Quantity	1
Wall count	2
Supports	-
Speed	Standard
Notes	-

BASE_light_mount_swivel



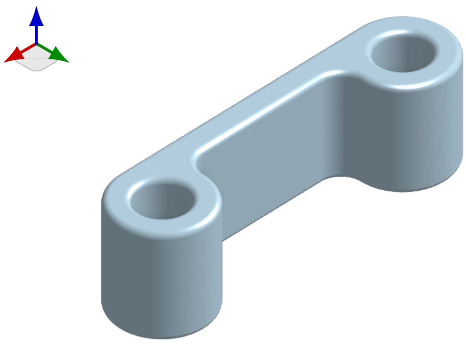
Quantity	1
Wall count	2
Supports	-
Speed	Standard
Notes	-

BASE_ruler



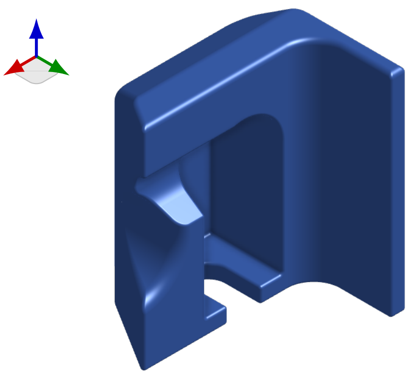
Quantity	1
Wall count	2
Supports	On highlighted faces only - Style: normal
Speed	Standard
Notes	-

BASE_top_cap



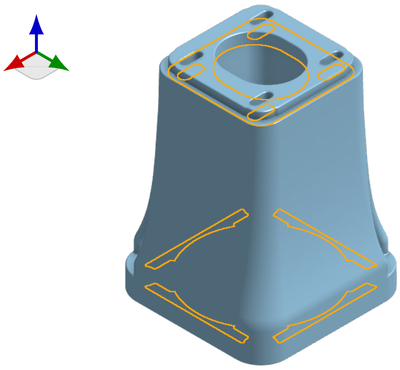
Quantity	1
Wall count	2
Supports	-
Speed	Standard
Notes	-

BASE_switch_housing



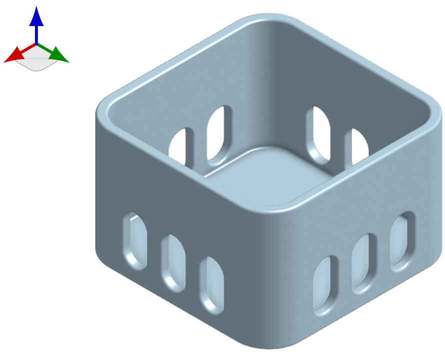
Quantity	1
Wall count	2
Supports	-
Speed	Standard
Notes	-

BLOCKS_camera_block



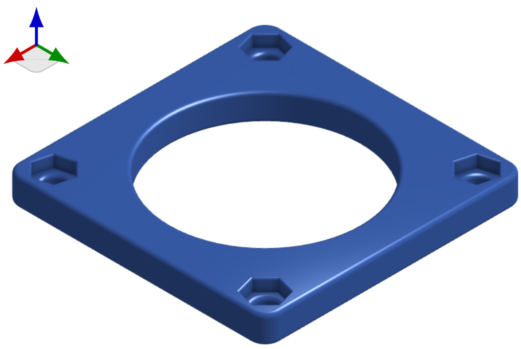
Quantity	1
Wall count	2
Supports	On highlighted faces only - Style: tree
Speed	Standard
Notes	-

BLOCKS_camera_cap



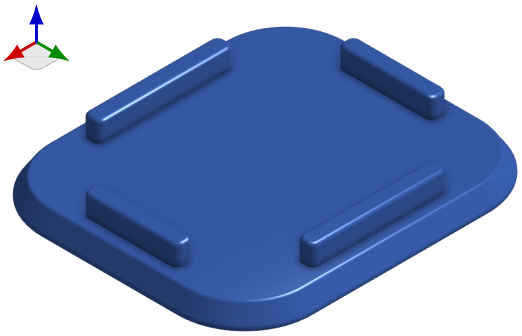
Quantity	1
Wall count	2
Supports	-
Speed	Standard
Notes	-

BLOCKS_camera_nut_holder



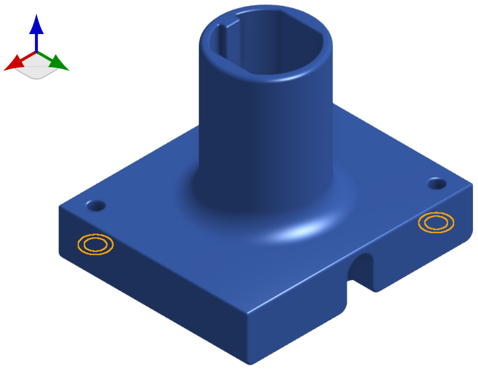
Quantity	1
Wall count	2
Supports	-
Speed	Standard
Notes	-

BLOCKS_cap



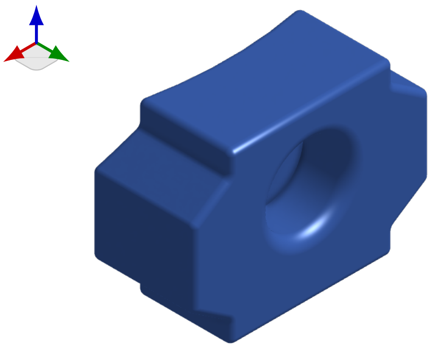
Quantity	2
Wall count	2
Supports	-
Speed	Standard
Notes	-

BLOCKS_detector_block



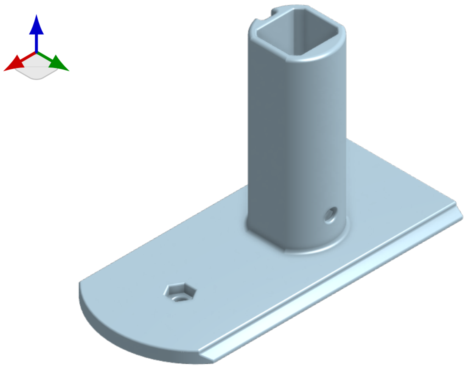
Quantity	1
Wall count	2
Supports	On highlighted faces only - Style: normal
Speed	Standard
Notes	-

BLOCKS_detector_clamp



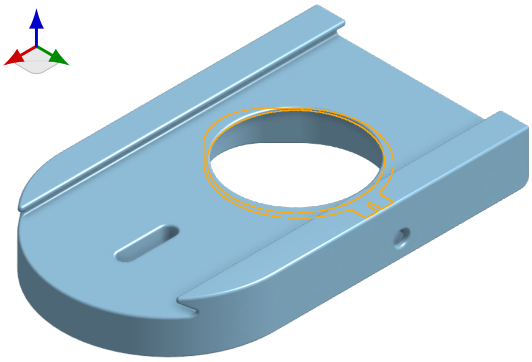
Quantity	1
Wall count	2
Supports	-
Speed	Standard
Notes	-

BLOCKS_detector_q_slider



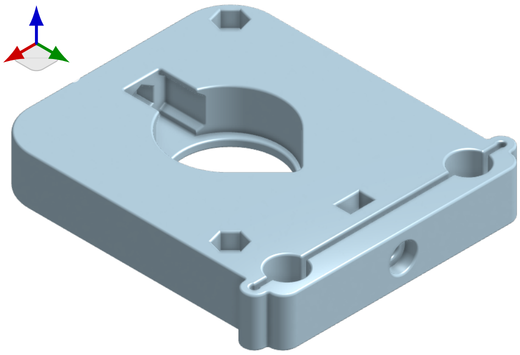
Quantity	1
Wall count	2
Supports	-
Speed	Standard
Notes	-

BLOCKS_detector_side_slider



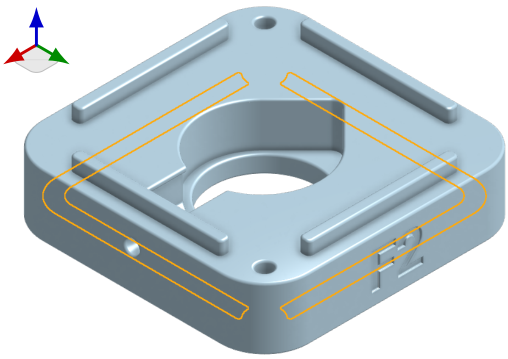
Quantity	1
Wall count	2
Supports	On highlighted faces only - Style: tree
Speed	Standard
Notes	-

BLOCKS_f1_block



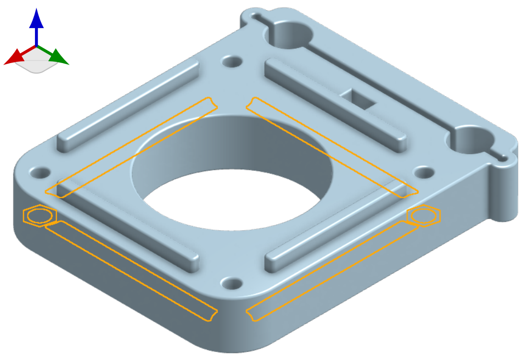
Quantity	1
Wall count	2
Supports	-
Speed	Standard
Notes	-

BLOCKS_f2_block



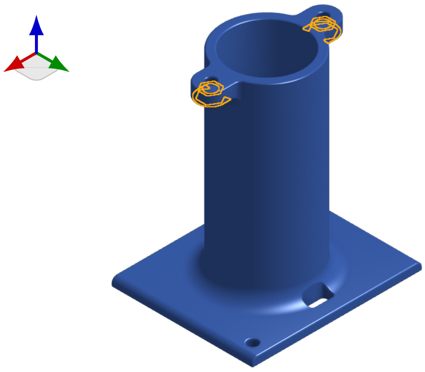
Quantity	1
Wall count	2
Supports	On highlighted faces only - Style: normal
Speed	Standard
Notes	-

BLOCKS_guide_spacer



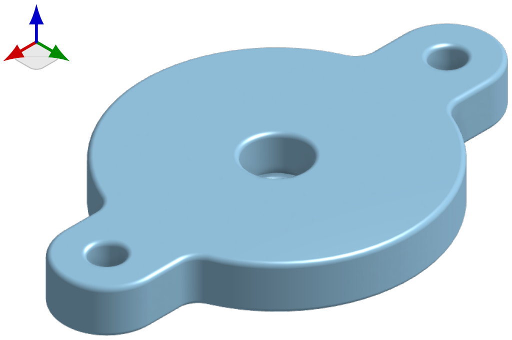
Quantity	1
Wall count	2
Supports	On highlighted faces only - Style: normal
Speed	Standard
Notes	-

BLOCKS_laser_block



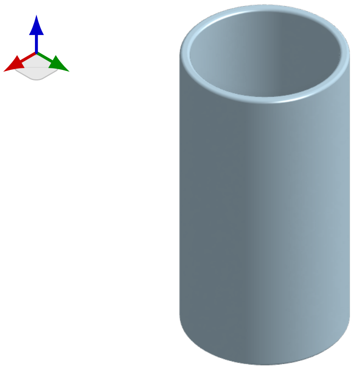
Quantity	1
Wall count	2
Supports	On highlighted faces only - Style: tree
Speed	Standard
Notes	-

BLOCKS_laser_cap



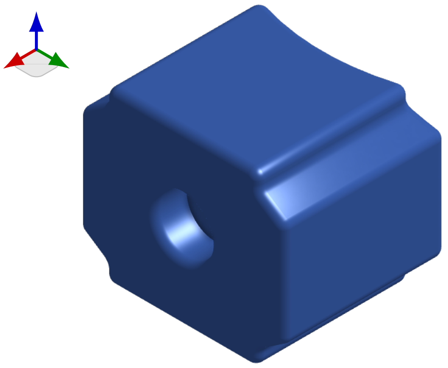
Quantity	1
Wall count	2
Supports	-
Speed	Standard
Notes	-

BLOCKS_laser_spacer



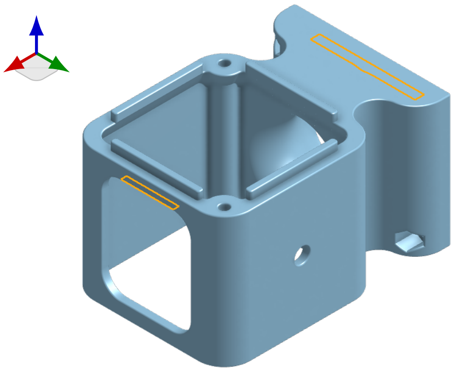
Quantity	1
Wall count	2
Supports	-
Speed	Standard
Notes	-

BLOCKS_lens_clamp



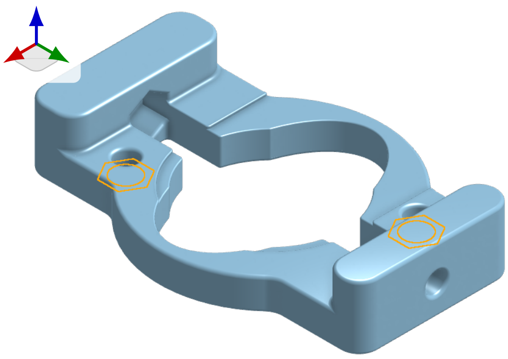
Quantity	2
Wall count	2
Supports	-
Speed	Standard
Notes	-

BLOCKS_mirror_block



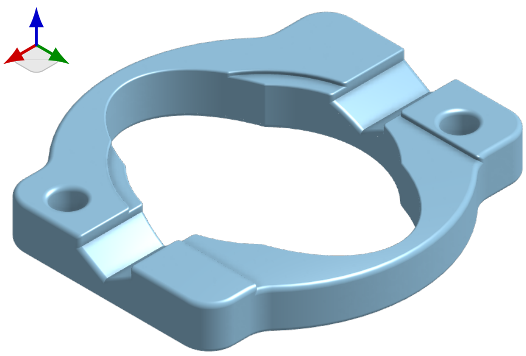
Quantity	2
Wall count	2
Supports	On highlighted faces only - Style: tree
Speed	Standard
Notes	-

BLOCKS_mirror_clamp_bottom



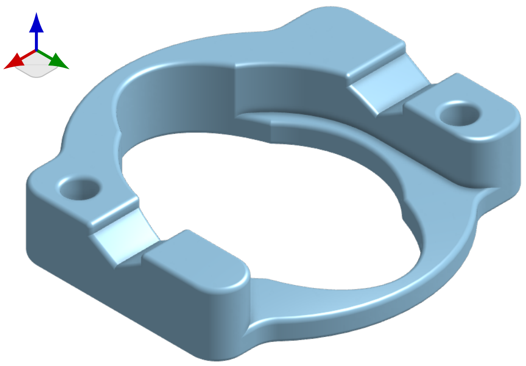
Quantity	2
Wall count	2
Supports	On highlighted faces only - Style: normal
Speed	Standard
Notes	-

BLOCKS_mirror_clamp_top



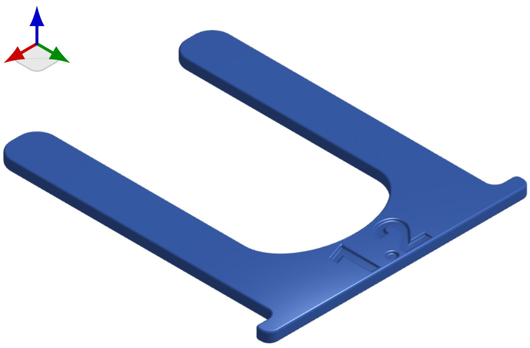
Quantity	1
Wall count	2
Supports	-
Speed	Standard
Notes	-

BLOCKS_redfilter_clamp_top



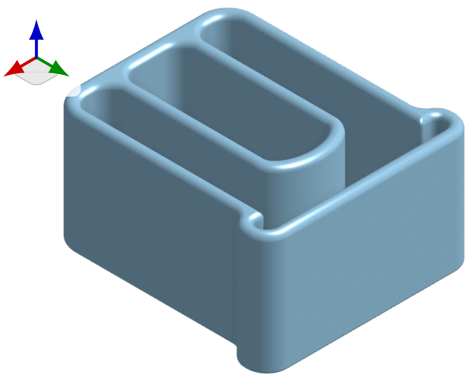
Quantity	1
Wall count	2
Supports	-
Speed	Standard
Notes	-

BLOCKS_spacers



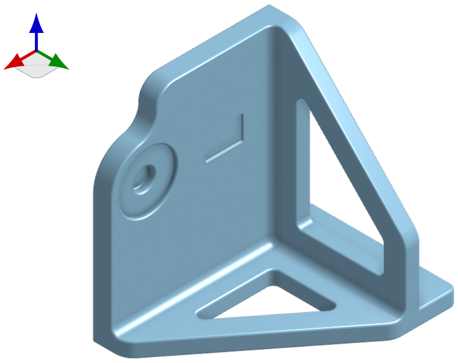
Quantity	8
Wall count	2
Supports	-
Speed	Standard
Notes	Print all individual spacers (from 0.4 to 3.2mm) with the same settings

BLOCKS_spacer_box



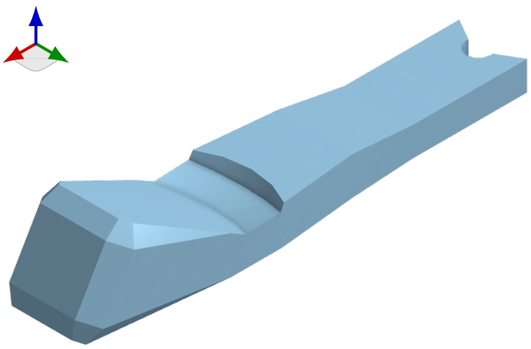
Quantity	1
Wall count	2
Supports	-
Speed	Standard
Notes	-

CALIBRATION_table_calibrator



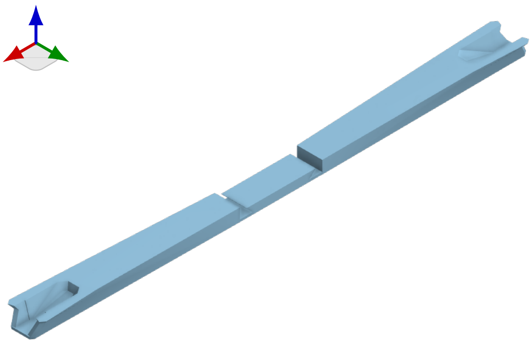
Quantity	1
Wall count	2
Supports	-
Speed	Standard
Notes	-

STAGE_actuator_nut_tool



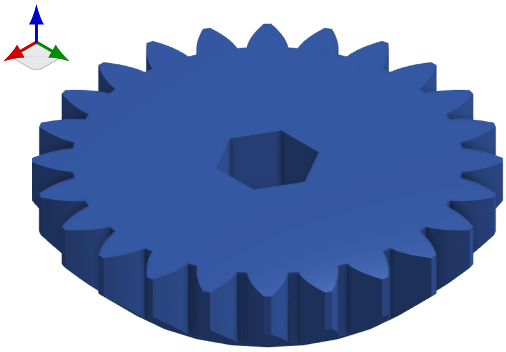
Quantity	1
Wall count	2
Supports	-
Speed	Walls: 40mm/s Infill: 80mm/s Bridges: 30mm/s
Notes	Printing settings copied from the OpenFlexure Block Stage assembly instructions

STAGE_actuator_ring_tool



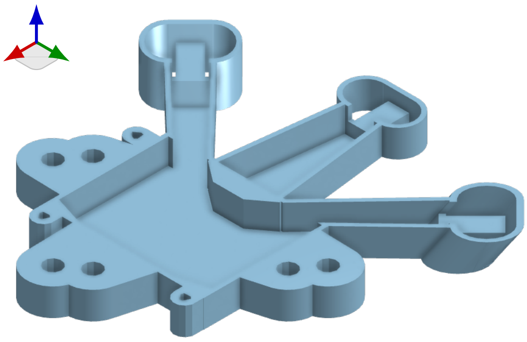
Quantity	1 needed, preferably 3 (breaks easily)
Wall count	2
Supports	-
Speed	Walls: 40mm/s Infill: 80mm/s Bridges: 30mm/s
Notes	Printing settings copied from the OpenFlexure Block Stage assembly instructions

STAGE_gear



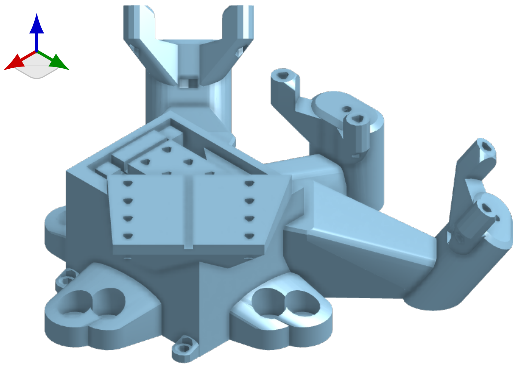
Quantity	3
Wall count	2
Supports	-
Speed	Walls: 40mm/s Infill: 80mm/s Bridges: 30mm/s
Notes	Printing settings copied from the OpenFlexure Block Stage assembly instructions

STAGE_main_base



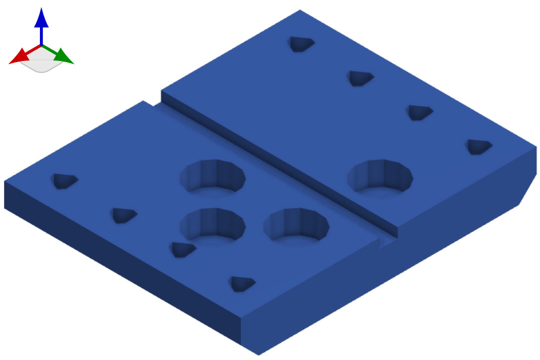
Quantity	1
Wall count	2
Supports	-
Speed	Walls: 40mm/s Infill: 80mm/s Bridges: 30mm/s
Notes	Printing settings copied from the OpenFlexure Block Stage assembly instructions

STAGE_main_body



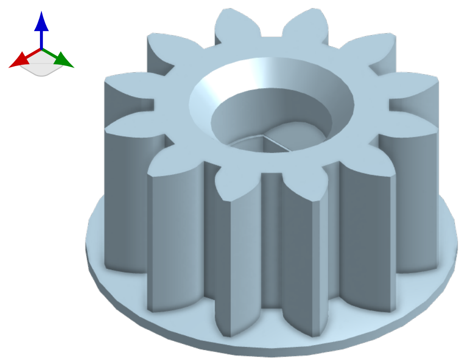
Quantity	1
Wall count	2
Supports	-
Speed	Walls: 40mm/s Infill: 80mm/s Bridges: 30mm/s
Notes	Printing settings copied from the OpenFlexure Block Stage assembly instructions

STAGE_moving_platform



Quantity	1
Wall count	2
Supports	-
Speed	Walls: 40mm/s Infill: 80mm/s Bridges: 30mm/s
Notes	Printing settings copied from the OpenFlexure Block Stage assembly instructions

STAGE_small_gear



Quantity	3
Wall count	2
Supports	-
Speed	Walls: 40mm/s Infill: 80mm/s Bridges: 30mm/s
Notes	Printing settings copied from the OpenFlexure Block Stage assembly instructions